

Bocheng Cui

bocheng.cui@unh.edu | +1 (603) 617 9302 | noahcui.github.io/info | github.com/noahcui

Publications

Anthony Westbrook, **Bocheng Cui**, Lakshmi Ongolu, Elizabeth Varki. “Active Reproducibility for File-Based Exploratory Science”, *Proceedings of the ACM Conference on Reproducibility and Replicability (ACM REP '26)*, 2026. [PDF]

Anthony Westbrook, Lakshmi Ongolu, **Bocheng Cui**, Elizabeth Varki. “Point-and-Click Reproducibility for Command-Line Data Analysis”, *Proceedings of the ACM Conference on Reproducibility and Replicability (ACM REP '26)*, 2026. [PDF]

Bocheng Cui, Aleksey Charapko. “Simpler is Better: Revisiting Mencius State Machine Replication”, *The 43rd International Symposium on Reliable Distributed Systems (SRDS)*, 2024.[PDF]

Owen Hilyard, **Bocheng Cui**, Marielle Webster, Abishek Bangalore Muralikrishna, Aleksey Charapko. “Cloudy Forecast: How Predictable is Communication Latency in the Cloud?”, *2024 33rd International Conference on Computer Communications and Networks (ICCCN)*, 2024. [PDF]

Technical Skills

Languages: Go, C/C++, Python, Java, SQL, R, Bash

Systems & Cloud: Docker, Kubernetes, OpenWhisk, Mininet, gRPC, AWS/GCP/Azure, Linux, Slurm, Git

AI-Assisted Development: Claude Code, GitHub Copilot; Anthropic/OpenAI API integration; prompt design for code generation, review, and testing

Domains: Distributed systems, cloud and serverless computing, computer networks, file systems

Experience

Software Development Engineer — Evolv Jun–Aug 2026

- Resolved incorrect 3D-to-2D camera projections by designing and implementing a pre-cut algorithm for virtual 3D imagery.
- Designed and implemented a fault-tolerant camera processing pipeline, achieving a modest improvement in frame rate (FPS).

Teaching Assistant — University of New Hampshire Aug 2021–Present

- Support instruction in undergraduate and graduate computer science courses.

Research Associate — UNH, NOAA-UNH Joint Hydrographic Center Jun–Aug 2024

- Developed a converter from Tiny 1.2 to the IHO SVG schema and migrated Python scripts from 2.7 to 3.6.

Research Assistant — University of New Hampshire May–Aug 2022

- Evaluated and optimized consistency protocols to address efficiency and reliability in distributed systems.

Cloud Software Developer — UNH, NOAA-UNH Joint Hydrographic Center May–Aug 2022

- Implemented and tested cloud-based software and architectures for hydrographic and bathymetric mapping.

Selected Projects

Provenance File System

- Designing and developing a provenance file system for cloud platforms to support tracking and management.

Mencius Revisited

- Studied Mencius, EPaxos, and Paxos; implemented improvements to Mencius’s failure-handling mechanisms that increased maximum throughput.

Fault-Tolerant Sharded Key/Value Storage

- Built a storage system using Raft and customizable shard placement; developed non-disruptive data migration and integrated Google MapReduce for data processing and analysis.

Cloud Latency Evaluation

- Built a latency-testing tool and evaluated communication latency and stability across AWS, GCP, and Azure.

Education

University of New Hampshire

Ph.D. in Computer Science, 2022–Present

University of New Hampshire

Master’s in Computer Science, 2018–2022

Qingdao University of Science & Technology

Bachelor’s in Material Science, 2014–2018